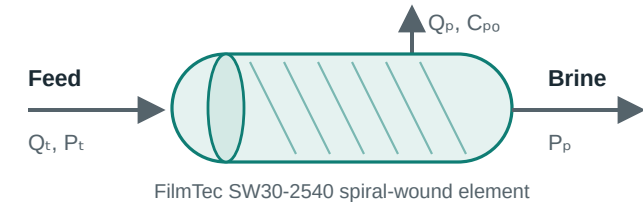


1 SETUP & DATA



Healthy
train sst+sss (47) · test sin (45)
generalization set

Degraded
test ssr (12)
anomaly-detection set

Task: map 2 inputs (Q_t, P_t) to 3 outputs (Q_p, C_{p0}, P_p); degradation appears almost only in permeate salinity C_{p0} .

2 MODELS & BENCHMARK

Screened 15 models — top 2 advance (App. A)



Physics-only and GPR lead on C_{p0} ; NN(=PINN) is the reference baseline.

PINN (= NN) — reference baseline
Physics loss has zero gradient on the network weights; predictions are **identical for $\lambda = 0, 1, 10, 100$** .

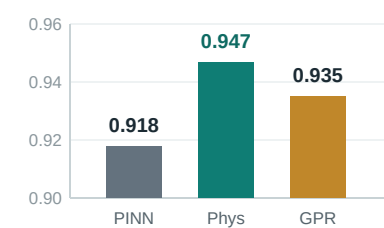
Physics-only regression best of 15
Network-free fit of the same six transport parameters ($L_p, B_s, \pi_0, k, n, \beta$).

Gaussian process regression
Adds predictive uncertainty $\hat{\sigma}$ — tested as a degradation indicator.

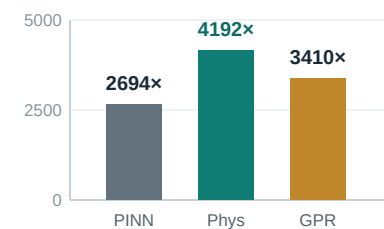
Hyperparameter sweep (App. B): no NN over learning rate, width, or depth (best 40.2) closes the gap with physics-only's 26.9 MSE.

3 RESULTS — PERMEATE SALINITY (C_{p0})

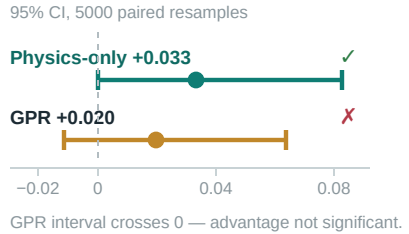
(a) Accuracy, R^2 on healthy test



(b) Anomaly ratio, ssr / sin MSE



(c) Bootstrap ΔR^2 vs PINN



- Interpretable β rises 7–15× when degraded — recovered with no network.
- GPR $\sigma \approx 7.4$, flat across healthy and degraded — misses the anomaly (×3410).